

Sri Lanka Institute of Information Technology



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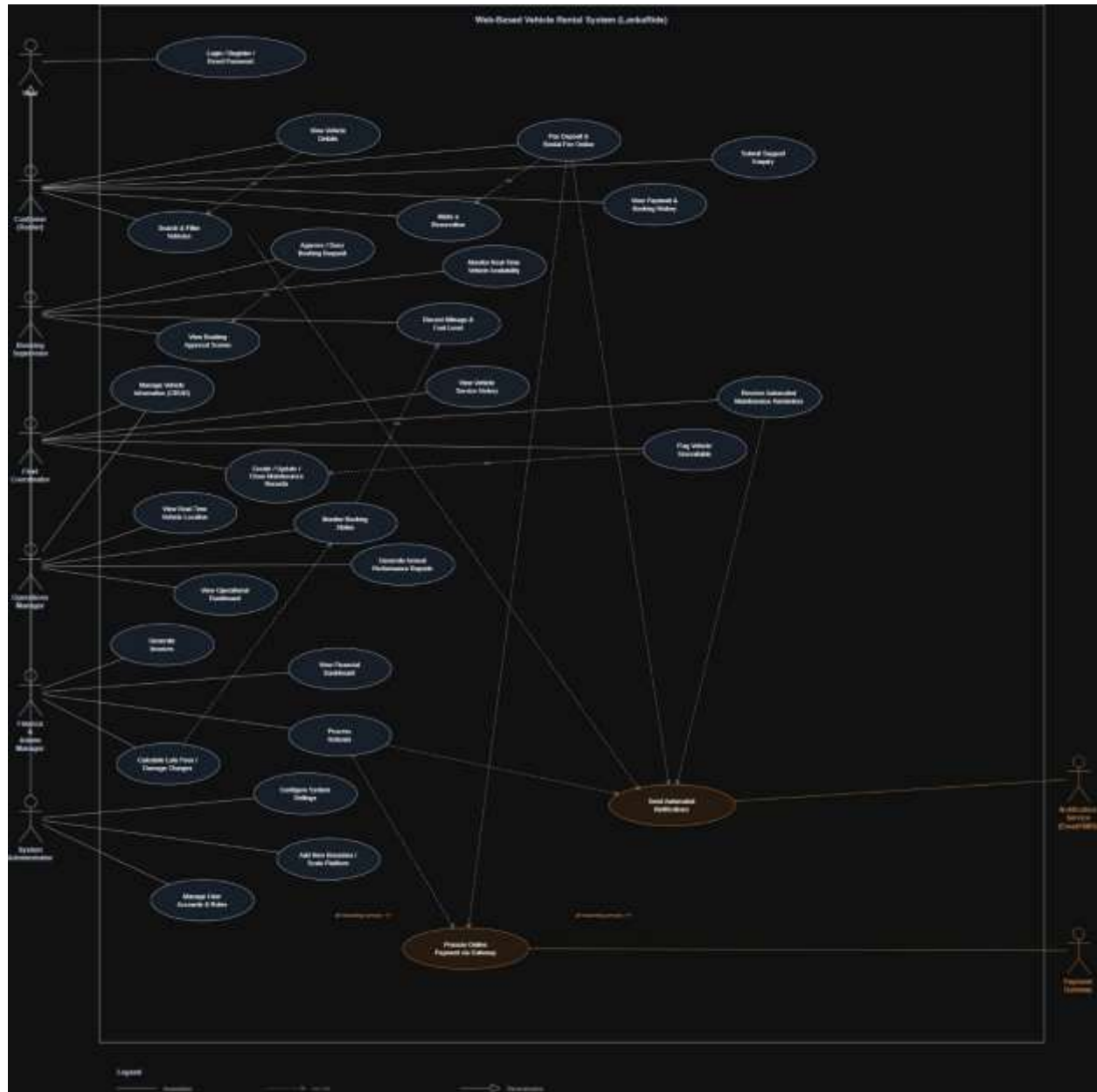
Use Case Diagram Assignment

Web-Based Vehicle Rental System – Lanka Ride Rentals

Group ID: 2026-Y2-S1-MLB-B13G2-05

IT24102871	PAHASARA Y.A.P
IT25102266	SAKALASOORIYA S.A.A.A
IT24101349	SAMARANAYAKE P.I.S
IT25102265	KAVINDI P.D.N
IT24101008	WICKRAMASINGHE R.D.W.K.G.S
IT25102264	DE SILVA D.L.K.C

1. Use Case Diagram



2. Individual Use Case Scenarios

Use case scenario 1: Create, Update and Close a Vehicle Maintenance/ Service Record
Group Member: DE SILVA D.L.K.C | Registration No: IT25102264

Use Case ID	UC - 01
Related Function	Fleet Maintenance & Servicing Management
Primary Actor	Fleet Coordinator
Secondary Actors	Booking & Reservation Module (system), Notification Service
Trigger	A vehicle is due for scheduled servicing, has a recorded fuel/milage discrepancy, or has a sustained damage

Description

This use case allows the Fleet Coordinator to log, update and close maintenance and service records for any vehicle in the fleet, keeping mileage, repair, cost and damage history accurate, and automatically excluding vehicles under maintenance from the booking pool so they cannot be double-booked.

Preconditions

The Fleet Coordinator is authenticated with the correct role, and the target vehicle already exists in the vehicle inventory.

Basic Flow of Events

1. Fleet Coordinator logs in and opens the Fleet Maintenance module.
2. Fleet Coordinator searches for the vehicle by registration number or selects it from the fleet listing.
3. System displays the vehicle's profile, current status and full service history.
4. Fleet Coordinator selects "Add Maintenance Record" and enters the service type (routine service, repair or damage assessment), date, estimated cost, description of work required, and the mechanic(s) assigned.
5. System validates the mandatory fields and saves the record against the vehicle's maintenance history.
6. System automatically triggers the "Flag Vehicle Unavailable" extension, moving the vehicle to the unavailability list so the Booking & Reservation module can no longer offer it, preventing double booking.
7. While servicing is underway, the Fleet Coordinator can update the record (status, revised completion date, parts replaced).

8. Once servicing is finished, the Fleet Coordinator closes the record with the final cost and completion date.
9. System automatically reinstates the vehicle's status to "Available" and notifies the Booking Supervisor via the Notification Service.

Extensions / Alternative Flows

- 4a. If the request originated from a mileage/fuel discrepancy flagged at vehicle return, the system pre-populates the description with the discrepancy note for review.
- 6a. If the vehicle already has confirmed future bookings, the system warns the Fleet Coordinator so those bookings can be reassigned or the customer informed before the vehicle is taken off the road.
- 8a. If further issues are discovered mid-service, the Fleet Coordinator extends the estimated completion date and the vehicle remains flagged unavailable.

Postconditions

The vehicle's maintenance history is complete and accurate, and its availability status correctly reflects whether it is fit for rental.

Use case scenario 2: Add, Update and Remove Vehicle Records

Group Member: SAKALASOORIYA S.A.A.A | Registration No: IT25102266

Use Case ID	UC - 02
Related Function	Vehicle Management
Primary Actor	Fleet Coordinator/ Operating Manager
Secondary Actors	Fleet Maintenance Module, Booking & Reservation Module
Trigger	A new vehicle joins the fleet, an existing vehicle's details change, or a vehicle is transferred or retired

Description

This use case covers full CRUD control of the vehicle inventory: creating new vehicle records, updating details, recording inter-branch transfers, and removing retired vehicles, so that every branch always has a single, synchronized source of truth for the fleet instead of the disconnected spreadsheets described in the project's problem statement.

Preconditions

The user holds Fleet Coordinator or Operations Manager privileges and is logged in.

Basic Flow of Events

1. User opens the Vehicle Management module.
2. User selects "Add New Vehicle", or selects an existing vehicle to update or remove.
3. User enters/edits the vehicle's category, brand, model, seating capacity, gearbox type, fuel type, features (e.g., GPS, air conditioning), photos, price per day, deposit amount and home branch.
4. System validates the fields (including a unique registration number) and saves the record to the central vehicle database, synchronized in real time across the Colombo, Kandy and Galle branches.
5. When a vehicle is physically relocated between branches, the user records the transfer; the system updates the "current location" field and timestamps the movement so staff at every branch immediately see the correct location.
6. If a vehicle is retired, the user selects "Remove Vehicle"; the system archives its historical booking and maintenance data and removes it from active search results.

Extensions / Alternative Flows

- 4a. If the vehicle is marked damaged or sent for maintenance, the system links to UC-01 and automatically places it on the unavailability list.
- 5a. If the vehicle has an active booking during a transfer, the system warns the user before the relocation is confirmed.

Postconditions

The fleet inventory across all branches is accurate and current, and every actor (customer, staff and management) sees the same real-time vehicle availability and location data.

Use case scenario 3: Make a Reservation and Approve/Deny a Booking Request

Group Member: SAMARANAYAKE P.I.S | Registration No: IT24101349

Use Case ID	UC - 03
Related Function	Reservation and Booking Management
Primary Actor	Customer (initiates); Booking Supervisor (approves)
Secondary Actors	Vehicle & Fleet Management module, Payment module, Notification Service
Trigger	A customer wants to rent a vehicle for specific dates and a branch

Description

This use case covers the full booking lifecycle: a customer searching for and reserving a vehicle, and a Booking Supervisor reviewing and approving or denying the request from a single consolidated screen, so bookings are processed quickly and double bookings across branches are prevented.

Preconditions

The customer is logged in (or browsing as a guest for search), and at least one vehicle matching the search criteria exists.

Basic Flow of Events

1. Customer searches and filters vehicles by type, price, gearbox, fuel type, branch and date range.
2. Customer selects a vehicle, reviews its details (photos, daily rate, deposit, extras) and chooses pickup/return dates and branch.
3. Customer submits the booking request; the system includes "Check Real-Time Availability" to confirm the vehicle is not already reserved or under maintenance for the selected dates.
4. System creates the booking with status "Pending" and forwards it to the Booking Supervisor at the relevant branch.
5. Booking Supervisor opens the "View Booking Approval Screen", where vehicle status, customer information, rental history and deposit/advance payment status are all visible together, without switching between screens.
6. Booking Supervisor approves or denies the request, optionally adding a reason.
7. System updates the booking status (Pending, Approved, Ongoing, Completed or Cancelled) and, through the Notification Service, informs the customer of the outcome by email/SMS.
8. If approved, the customer proceeds to pay the deposit and rental fee online (see UC-04).

Extensions / Alternative Flows

- 3a. If the vehicle becomes unavailable between search and submission (e.g., booked by another customer), the system rejects the request immediately and suggests similar alternatives.
- 6a. If denied, the Booking Supervisor must select a reason (e.g., incomplete documentation, vehicle unavailable) that is sent to the customer.
- 7a. The Operations Manager can, at any time, monitor the status of all bookings across branches through a separate read-only dashboard view.

Postconditions

A confirmed booking record exists with a definite status, and the vehicle's real-time availability is updated so it cannot be double-booked at another branch.

Use case scenario 4: Pay Booking Deposit/ Rental Fee, Generate Invoice and Process Refund

Group Member: KAVINDI P.D.N | Registration No: IT25102265

Use Case ID	UC - 04
Related Function	Payment & Billing Management
Primary Actor	Customer (payment); Finance & Admin Manager(invoicing/refunds)
Secondary Actors	Payment Gateway (external, sandbox/test), Notification Service
Trigger	A customer's booking is approved and awaiting payment, a vehicle is returned, or a cancellation is authorized

Description

This use case manages the financial side of a rental: secure online payment of deposits and rental fees, automatic invoice generation, late-fee/damage-charge calculation on return, and refund processing for authorized cancellations, giving both the customer and the Finance & Admin Manager a transparent, auditable transaction trail.

Preconditions

The customer has an approved booking, and the Payment Gateway sandbox/test service is reachable.

Basic Flow of Events

1. After booking approval, the customer is prompted to pay the deposit and rental fee online.
2. Customer selects a payment method and confirms the amount, less any registered-customer discount earned from bookings placed during the month, where applicable.
3. System includes "Process Online Payment via Gateway", forwarding the transaction securely to the sandbox payment gateway.
4. On success, the gateway confirms the transaction; the system marks the booking's payment status as "Paid" and includes "Send Automated Notifications" to email the customer a receipt.

5. System automatically generates an invoice for the completed transaction and stores it in the customer's payment history.
6. When the vehicle is returned, the Finance & Admin Manager reviews the recorded mileage/fuel data and the system calculates any applicable late fees or damage charges as a supplementary invoice line.
7. If a booking is cancelled within the refund policy window, the Finance & Admin Manager processes the refund; the system includes the Payment Gateway again to reverse the transaction and notifies the customer once completed.

Extensions / Alternative Flows

- 3a. If the payment gateway declines the transaction, the system shows the reason and lets the customer retry with another method; the booking stays "Pending Payment".
- 6a. If no late fee or damage is found on return, this step is skipped and the booking is simply marked "Completed".
- 7a. If a cancellation falls outside the refund policy, the Finance & Admin Manager may issue a partial refund with a documented justification.

Postconditions

All financial transactions — payments, invoices, refunds and late fees — are accurately recorded with a visible audit trail, and both the customer and the Finance & Admin Manager have up-to-date visibility of the transaction status.

Use case scenario 5: View Dashboard and Generate Daily/ Monthly/ Annual Reports

Group Member: Wickramasinghe R.D.W.K.G.S | Registration No: IT24101008

Use Case ID	UC - 05
Related Function	Reporting and Admin Dashboard
Primary Actor	Operations Manager/ Finance & Admin Manager
Secondary Actors	None (reads directly from the central database)
Trigger	A manager needs an up to date operational or financial view for decision making

Description

This use case gives management real-time visibility of operational and financial performance through a single dashboard, replacing manual, spreadsheet-based report compilation with automatically generated daily, monthly and annual reports.

Preconditions

The user is authenticated as an operations Manager or Finance & Admin Manager, and booking, payment and fleet data exist for the requested period.

Basic Flow of Events

1. User opens the Admin Dashboard.
2. System retrieves and displays key real-time metrics: rentals in progress, transactions received, deposits collected, refunds issued and total income for the day.
3. User selects a reporting period: Daily, Monthly or Annual.
4. For a Monthly report, the system compiles total revenue, bookings, outstanding payments, refunds and the most-rented vehicle categories.
5. For an Annual report, the system compiles overall performance, profit/revenue trends, a branch-based comparison (Colombo, Kandy, Galle) and vehicle utilization rates.
6. User can filter the report by branch, vehicle category or date range.
7. System renders the results as charts and tables on the dashboard.
8. User exports or prints the report for a management meeting.

Extensions / Alternative Flows

- 3a. If no data exists for the selected period (e.g., a newly opened branch), the system shows an empty-state message instead of an error.
- 6a. If a filter combination returns no records, the system suggests broadening the filter.

Postconditions

Management has an accurate, current view of operational and financial performance, supporting informed decisions without manual report compilation.

Use case scenario 6: Send Automated Notifications and Manage Customer Support Enquiries

Group Member: PAHASARA Y.A.P | Registration No: IT24102871

Use Case ID	UC - 06
Related Function	Customer Service & Alerts
Primary Actor	Customer (raises enquiry); System (sends notifications)
Secondary Actors	Notification Service (Email/SMS gateway), Booking Supervisor/ Customer Service Staff
Trigger	A booking/ payment/ maintenance event occurs, or a customer has an enquiry

Preconditions

A triggering event has occurred elsewhere in the system (booking approval, payment confirmation, maintenance reminder), or a logged-in customer wants to raise an enquiry.

Basic Flow of Events (Automated Notifications)

1. Another module (Booking, Payment or Fleet Maintenance) completes an action that requires the customer or staff to be informed, e.g. booking approved, payment received, or a service reminder is due.
2. That module includes "Send Automated Notifications".
3. System composes the appropriate message (booking confirmation, payment receipt, or reminder) and sends it via email and/or SMS through the Notification Service.
4. System logs the notification against the relevant booking/customer record for traceability.

Extensions / Alternative Flows

- 3a. If the Notification Service is temporarily unreachable, the system queues the message and retries automatically.
- 6a. Urgent enquiries linked to an ongoing rental are automatically prioritized for faster staff response.
- Basic Flow (Support Enquiries) — 5. Customer opens "Submit Support Enquiry" and describes the issue, optionally linking it to a booking. 6. System creates a ticket with status "Open" and assigns it to the relevant branch's customer service staff. 7. Staff reviews and responds, updating the status (In Progress, Resolved, Closed). 8. Customer is notified automatically when the ticket receives a response or is resolved.

Postconditions

Customers remain informed at every key stage of the rental lifecycle, and every support enquiry is tracked to resolution with a full history.